

PREREGISTRATION — dark-matter-quantum-sim

Series: dark-matter-quantum-sim

Steward: Remnant Fieldworks Inc.

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Status: LOCKED (hash recorded in MANIFEST.sha256)

0. Covenant

This document is written and locked **before** any results are computed. It follows the Remnant Fieldworks preregistration covenant used across the prior 75 preregistered quantum experiments (WITNESS, BELLWETHER, CHRONO, OMNI, TRINITY series):

1. **Questions and thresholds are stated first.** No result may retroactively change the question it was meant to answer.
2. **Every run emits a ProofRecord** bound by SHA-256 to its own contents.
3. **Verdicts are** PASS / FAIL / HOLD against thresholds fixed in this document.
4. **Publish regardless of verdict.** A FAIL is published with the same discipline as a PASS. Suppression of a negative result is a covenant violation.

Once the SHA-256 of this file is written to MANIFEST.sha256, the questions, Hamiltonians, observables, and thresholds below are considered frozen for version v1.

1. HONEST SCOPE (binding)

These are **small-system toy Hamiltonian simulations** (2–4 qubits) that model the mathematical structure of dark matter candidate interactions. They are:

- **NOT** detection experiments,
- **NOT** simulations of real dark matter,
- and they **do NOT** prove that any dark matter candidate particle is real.

The value of this work is **methodological**:

1. Applying preregistered IBM Q simulation methods to open physics questions.
2. Binding simulation results to ExecutionProof-style ProofRecords using the same framework as the Remnant Fieldworks quantum witness series.
3. Establishing prior art in quantum-simulation-as-governed-research.

A physics collaborator is required to develop more realistic Hamiltonians beyond these toy models. Nothing in this repository should be read as a physical discovery claim.

2. Experiments

Three experiments are preregistered. For each we fix: the **question**, the **Hamiltonian**, the **observable**, and the **pass/fail threshold**. Conventions:

- Pauli operators $\sigma_x, \sigma_y, \sigma_z$; identity I . Tensor order is $\text{qubit0} \otimes \text{qubit1}$.
- Natural units, $\hbar = 1$. Time t is dimensionless in simulation units unless stated.
- Classical simulation uses exact matrix exponentiation $U(t) = \exp(-i H t)$.

EXPERIMENT DM-001 — Axion Dark Matter Toy Model

Question. Does a 2-qubit anharmonic-oscillator approximation of the Peccei–Quinn double-well potential produce a coherent oscillation in $\langle \sigma_z \rangle$ with the expected period?

Background. The QCD axion (Peccei–Quinn, 1977) is a hypothetical particle proposed to solve the strong CP problem. Its mass–coupling relationship yields a characteristic oscillation signature. On a quantum computer the axion potential can be approximated as an anharmonic oscillator.

Hamiltonian.

$$H = \omega_0 (\sigma_z \otimes I + I \otimes \sigma_z) + \lambda (\sigma_x \otimes \sigma_x)$$

with axion frequency parameter $\omega_0 = 1.0$ and coupling $\lambda = 0.3$.

Observable. Expectation value $\langle \sigma_z \rangle$ on qubit 0 as a function of time (oscillation signature), initial state $|00\rangle$.

Classical method. Matrix exponentiation.

IBM Q circuit. R_y rotations + CNOT, measured in the Z basis (Trotterized time evolution).

Preregistered threshold.

- Classical: the dominant oscillation period T of $\langle \sigma_z \rangle$ satisfies $T = 2\pi/\omega_0 \pm 5\%$.
- IBM Q: the hardware result matches the classical prediction within 2σ (accounting for hardware noise).
- Verdict **PASS** iff the classical period is within $\pm 5\%$ of $2\pi/\omega_0$.

EXPERIMENT DM-002 — Sterile Neutrino Oscillation Toy Model

Question. Does a 2-qubit Trotterized evolution reproduce the analytic 2-flavor neutrino survival probability for a maximal-mixing (sterile-neutrino-analogue) scenario?

Background. Sterile neutrinos are dark matter candidates that mix with active neutrinos. Neutrino oscillation is already fully described by quantum mechanics (PMNS matrix), so a 2-qubit system naturally models 2-flavor oscillation. This is the **most faithfully simulatable** of the three experiments.

Hamiltonian.

$$H = (\Delta m^2 / 4E) (-\cos(2\theta) \sigma_z + \sin(2\theta) \sigma_x)$$

Initial state $|\nu_\mu\rangle = \cos(\theta)|0\rangle + \sin(\theta)|1\rangle$.

Parameters: mixing angle $\theta = \pi/4$ (maximal mixing), $\Delta m^2 = 1 \text{ eV}^2$
(light sterile-neutrino candidate range, LSND/MiniBooNE anomaly band).

Observable. Survival probability

$$P(\nu_\mu \rightarrow \nu_\mu) = 1 - \sin^2(2\theta) \cdot \sin^2(1.27 \Delta m^2 L / E)$$

Classical method. Exact analytic formula (and matrix-exponentiation cross-check).

IBM Q circuit. $R_y(2\theta)$ state prep + Trotterized time evolution.

Preregistered threshold.

- IBM Q survival probability matches the analytic prediction within **10%** (loose — deep circuits accumulate hardware noise).
- Verdict **PASS** iff the classical/analytic and matrix-exponentiation curves agree within 1% at all sampled points (kill condition, §3) **and** the reported point matches analytic within 10%.

EXPERIMENT DM-003 — WIMP Scattering Toy Model (Simplified)

Question. Does a 2-qubit $XX+YY$ (exchange) coupling reproduce the expected $\sin^2(gt)$ spin-flip probability for a simplified WIMP-nucleon scattering analogue?

Background. WIMPs (Weakly Interacting Massive Particles) scatter off nucleons via the weak force; the cross-section sets detection rates. A 2-qubit system models a simplified WIMP-nucleon interaction as a spin-spin coupling.

Hamiltonian.

$$H = g (\sigma_x \otimes \sigma_x + \sigma_y \otimes \sigma_y)$$

qubit 0 = WIMP spin, qubit 1 = nucleon spin. Initial state $|\uparrow\downarrow\rangle = |01\rangle$.
Coupling $g = 1.0$ (scanned over a range for the rate-vs-coupling curve).

Observable. Probability of spin-flip to $|\downarrow\uparrow\rangle = |10\rangle$ as a function of g and t .

Classical method. Exact matrix exponentiation.

IBM Q circuit. Bell-state / exchange rotation circuit.

Preregistered threshold.

- Spin-flip probability matches $P = \sin^2(2gt)$ for the $XX+YY$ convention at the sampled times; specifically at $t = \pi/(4g)$ (a quarter of the $2g$ Rabi period) the flip probability equals $\sin^2(\pi/2) = 1$ within **±5%**.
- Verdict **PASS** iff the simulated flip probability matches the closed form within **±5%**.

Convention note (frozen here before results): for $H = g(XX+YY)$ acting on the single-excitation subspace $\{|01\rangle, |10\rangle\}$, the effective two-level gap is $2g$, giving $P_{\text{flip}}(t) = \sin^2(2gt)$. The quarter-period is therefore $t = \pi/(4g)$. This closed form is the reference used for the **±5%** test.

3. Kill condition (applies to all experiments)

If the **classical matrix-exponentiation simulation disagrees with the closed-form / analytic solution by more than 1%** at any sampled point, the classical layer itself is untrustworthy. In that case:

- the run is marked **HOLD** ,
- no **PASS / FAIL** verdict on the physics question is issued,
- the discrepancy is investigated before any IBM Q run is attempted.

This is a stop condition, not a failure of the physics question.

4. Publication rule

Results are published **regardless of PASS / FAIL / HOLD** , with the ProofRecord and its SHA-256 hash, to the Remnant Fieldworks Zenodo record for this series. A negative or null result is a valid, publishable outcome and is reported with identical prominence to a positive one.

5. Freeze

Upon writing `sha256(PREREGISTRATION.md)` to `MANIFEST.sha256` , sections 1–4 above are frozen for version `v1` of the `dark-matter-quantum-sim` series. Any change to a question, Hamiltonian, observable, threshold, or kill condition requires a new preregistration (`v2`) with a new hash and a new dated entry.